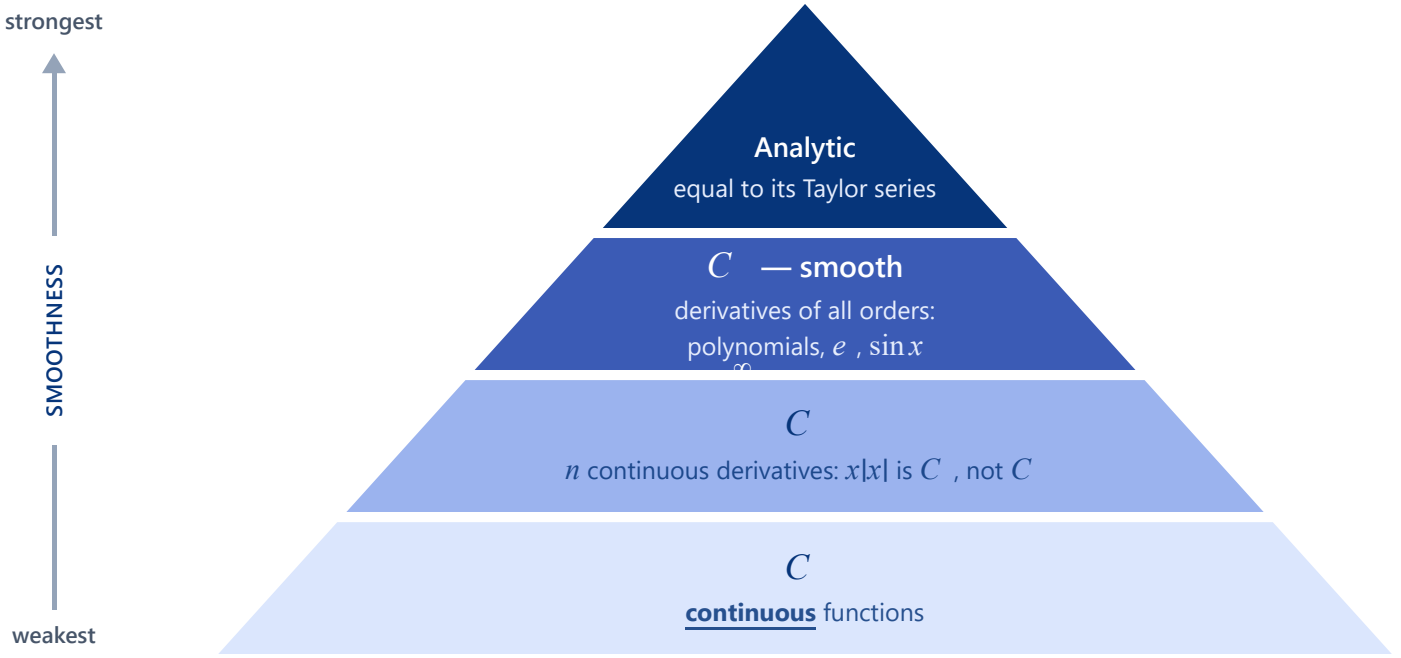
BEFORE YOU READ

## Classes by number of continuous derivatives

A function is in $C^n$ when $f, f', \ldots, f^{(n)}$ all exist and are continuous. Each level up asks for more, so every class sits inside the one below it.




TAKEAWAY

## Smooth is not the top

Analytic is strictly stronger than $C^\infty$: $e^{-1/x^2}$ (with $f(0) = 0$) is smooth, yet its Taylor series at $0$ is identically zero and does not represent it.